

Courses & Certifications Guide

Practice dataset for Project 3 — AI Course & Certification Advisor RAG

Important: This is a fictional practice dataset created for learning purposes. The courses, durations, prerequisites, and certification details are not official provider information.

1. Python Programming Fundamentals

Basic Python syntax, variables, functions, data structures, file handling, exceptions, and object-oriented programming. Prerequisite: basic computer programming knowledge. Duration: 8 weeks. Difficulty: Beginner. Skills: Python programming, problem solving, debugging, and basic software development. Career relevance: software development, automation, data analysis, and AI learning.

2. Data Analytics with Python

Python for data cleaning, analysis, visualization, and introductory statistics. Prerequisite: basic Python programming. Duration: 10 weeks. Difficulty: Intermediate. Skills: Pandas, NumPy, visualization, exploratory data analysis, and basic statistics. Career relevance: data analyst and junior data science roles.

3. Machine Learning Foundations

Supervised learning, unsupervised learning, model evaluation, feature preparation, and common machine learning algorithms. Prerequisites: Python programming and basic statistics. Duration: 12 weeks. Difficulty: Intermediate. Skills: machine learning workflow, model training, evaluation, and feature engineering. Career relevance: junior machine learning and AI roles.

4. Generative AI Application Development

Large language models, prompting, embeddings, retrieval-augmented generation, vector databases, and AI application development. Prerequisites: basic Python and familiarity with APIs. Duration: 10 weeks. Difficulty: Intermediate. Skills: LLM APIs, prompt engineering, embeddings, RAG, vector search, and AI application design. Career relevance: GenAI application development and AI engineering.

5. Cloud Computing Fundamentals

Cloud concepts, compute, storage, networking, databases, identity, monitoring, and deployment basics. Prerequisites: basic computer and networking knowledge. Duration: 8 weeks. Difficulty: Beginner to Intermediate. Skills: cloud architecture, virtual machines, storage, networking, and deployment fundamentals. Career relevance: cloud support, cloud engineering, DevOps, and software deployment.

6. AWS Certified Cloud Practitioner

Foundational AWS cloud concepts, core services, security, pricing, and architecture. Prerequisite: no prior AWS experience required. Preparation: 6 weeks. Difficulty: Beginner. Skills: AWS service awareness, cloud terminology, security fundamentals, and pricing concepts. Career relevance: starting certification for a cloud career. Exam: AWS Certified Cloud Practitioner examination.

7. Microsoft Azure Fundamentals (AZ-900)

Cloud concepts, Azure architecture and services, management, governance, and basic security.

Prerequisite: no advanced technical prerequisites. Preparation: 6 weeks. Difficulty: Beginner. Skills: Azure service awareness, cloud concepts, governance, and basic cloud security. Career relevance: Azure and cloud-focused careers. Exam: AZ-900.

8. Google Cloud Digital Leader

Digital transformation, cloud concepts, Google Cloud services, data, AI, security, and business value.

Prerequisite: no programming experience required. Preparation: 5 weeks. Difficulty: Beginner. Skills: cloud concepts, Google Cloud service awareness, data and AI concepts, and cloud business understanding.

Career relevance: foundational cloud knowledge without a programming-heavy start. Exam: Google Cloud Digital Leader.

9. Docker and Container Fundamentals

Containers, images, Dockerfiles, networking, volumes, and basic deployment. Prerequisites: basic Linux or command-line knowledge and programming familiarity. Duration: 5 weeks. Difficulty: Intermediate.

Skills: Docker image creation, container management, networking, and application packaging. Career relevance: software development, DevOps, cloud, and deployment roles.

10. DevOps Foundations

Version control, CI/CD, automated testing, containerization, deployment pipelines, and DevOps practices.

Prerequisites: basic programming and Git knowledge. Duration: 10 weeks. Difficulty: Intermediate. Skills: Git workflows, CI/CD, automation, Docker basics, and deployment practices. Career relevance: DevOps, cloud engineering, and software delivery.